SECTION 1 (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated **according to the following marking scheme**:

Full Marks : $+3$ If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

Q.1 Consider the function $f:(0,\infty)\to(-\infty,\infty)$ given by

$$f(x)=\sqrt{x}\,\log_e(x)-x+1\,.$$

Then which one of the following statements is TRUE ?

(A)	The derivative of the function f is decreasing in the interval $(0,1)$
(B)	The function f has a local maximum at some point $a\in(0,\infty)$
(C)	The function f has a local minimum at some point $b\in(0,\infty)$
(D)	The function f has **NEITHER** a point of local maximum **NOR** a point of local minimum in the interval $(0,\infty)$

Q.2 Let P be the point on the parabola $y=x^2$ such that the slope of the tangent to the parabola at the point P is 4. Let Q be the point in the first quadrant lying on the circle $x^2+y^2=2$ such that the slope of the tangent to the circle at the point Q is -1. Let R be the point in the first quadrant lying on the ellipse $x^2+4y^2=8$ such that the slope of the tangent to the ellipse at the point R is $-\frac{1}{2}$. Then the radius of the circle passing through the points P,Q and R is

(A)	$\sqrt{10}$	(B)	$\sqrt{5}$	(C)	$\sqrt{\frac{5}{2}}$	(D)	$2\sqrt{5}$

Q.3 Which one of the following matrices can be obtained by performing elementary row transformations on the 3×3 identity matrix ?

(A)	$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$	(B)	$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 1 & 2 & 1 \end{bmatrix}$
(C)	$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 2 & 5 & 8 \end{bmatrix}$	(D)	$\begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 2 \\ 0 & 2 & 3 \end{bmatrix}$

Q.4 Considering only the principal values of the inverse trigonometric functions, the value of

$$\cot^{-1}(\cot(-11)) + 10 \sin\left(2\cos^{-1}\left(\frac{1}{\sqrt{2}}\right)\right) + 10\sin(2\tan^{-1}(2))$$

is

(A)	$3\pi + 7$	(B)	7	(C)	$4\pi + 7$	(D)	$3\pi - 5$

SECTION 2 (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated **according to the following marking scheme**:

  Full Marks : $+4$ **ONLY** if (all) the correct option(s) is(are) chosen;
  Partial Marks : $+3$ If all the four options are correct but **ONLY** three options are chosen;
  Partial Marks : $+2$ If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
  Partial Marks : $+1$ If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : -1 In all other cases.
- For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then
  choosing ONLY (A), (B) and (D) will get $+4$ marks;
  choosing ONLY (A) and (B) will get $+2$ marks;
  choosing ONLY (A) and (D) will get $+2$ marks;
  choosing ONLY (B) and (D) will get $+2$ marks;
  choosing ONLY (A) will get +1 mark;
  choosing ONLY (B) will get +1 mark;
  choosing ONLY (D) will get +1 mark;
  choosing no option (i.e. the question is unanswered) will get 0 marks; and
  choosing any other combination of options will get -1 marks.

Q.5 Suppose that Box I contains 6 red balls and 9 green balls, and Box II contains 8 red balls and 12 green balls. All the balls of Box I and Box II are mixed together and a ball is chosen at random from them. Let E_1 be the event that the ball chosen belonged to Box I and let E_2 be the event that the ball chosen belonged to Box II. Let F_1 be the event that the ball chosen is red and let F_2 be the event that the ball chosen is green.

Then which of the following statements is (are) TRUE ?

(A)	The events E_1 and F_1 are independent
(B)	The events E_2 and F_2 are dependent
(C)	The conditional probability $P(F_1 \mid E_1)$ is equal to the conditional probability $P(F_1 \mid E_2)$
(D)	The conditional probability $P(F_1 \mid E_1)$ is greater than the conditional probability $P(F_2 \mid E_2)$

Q.6

Let P be the plane such that it contains the straight line $\frac{x-1}{2}=\frac{y-3}{3}=\frac{z+2}{1}$ and is perpendicular to the plane $x+2y+3z=4$. Let P_1 be the plane which passes through the point $(4,2,2)$ and is parallel to P.

Then which of the following statements is (are) TRUE ?

(A)	The equation of the plane P is $7x-5y+z=-10$
(B)	The distance between the planes P and P_1 is 30
(C)	The distance of the plane P from the origin is $2\sqrt{3}$
(D)	The acute angle between the plane P and the plane $2x+2y+z=3$ is $\cos^{-1}\left(\frac{1}{3\sqrt{3}}\right)$

Q.7

Let $\mathbb{R}$ denote the set of all real numbers. Let $f:\mathbb{R}\to\mathbb{R}$ be an arbitrary function and let $g:\mathbb{R}\to\mathbb{R}$ be the function defined by

$$g(x)=xf(x), \text{ for all } x\in\mathbb{R}.$$

Then which of the following statements is (are) TRUE ?

(A)	The function g is always continuous at $x=0$
(B)	If f is continuous at $x=0$, then g is differentiable at $x=0$
(C)	If g is differentiable at $x=0$, then f is continuous at $x=0$
(D)	If g is differentiable at $x=0$, then $\lim_{x\to 0} f(x)$ exists

Q.8

Consider the matrix

$$M = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}.$$

Let p , q , r , s , a , b , c and d be integers such that

$$M^{26} = \begin{bmatrix} p & q \\ r & s \end{bmatrix} \quad \text{and} \quad \sum_{k=1}^{26} M^k = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Then which of the following statements is (are) TRUE ?

(A)	There exists a 2×2 invertible matrix N with real entries such that $MN = N\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
(B)	The value of a is 378
(C)	For any two given integers m and n, there exist unique integers x and y such that $px + qy = m \quad \text{and} \quad rx + sy = n$
(D)	For each positive real number t, the system of linear equations $(a + t)x + by = 1$ $cx + (d + t)y = -1$ has a unique solution

SECTION 3 (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme**:
  Full Marks : $+4$ If **ONLY** the correct numerical value is entered in the designated place;
  Zero Marks : 0 In all other cases.

Q.9 Let $S = \{1, 2, 3, \ldots, 10\}$. Consider the set

$$X = \{R : R \text{ is an equivalence relation on the set } S \text{ such that } R \text{ has exactly } 42 \text{ elements}\}.$$

Then the number of elements in X is _____.

Q.10 Consider the function $f : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to (-\infty, \infty)$ defined by

$$f(x) = (|x| + |x-1|)\sin x + [x \sin x],$$

where $[x \sin x]$ is the greatest integer less than or equal to $x \sin x$.

Let α be the total number of points in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ at which f is **NOT** continuous, and let β be the total number of points in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ at which f is **NOT** differentiable.

Then the value of $\alpha + \beta$ is _____.

Q.11 The number of ways to distribute 10 identical red pens and 14 identical blue pens among four persons such that each person gets 6 pens, is _____.

Q.12 Let

$$\alpha = \left(1 - 2\cos\left(\frac{\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{3\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{9\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{27\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{81\pi}{11}\right)\right).$$

Then the value of $5 - \alpha^2$ is _____.

SECTION 4 (Maximum Marks: 16)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated **according to the following marking scheme**:
  Full Marks : $+4$ **ONLY** if the option corresponding to the correct combination is chosen;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : -1 In all other cases.

Q.13

Match each entry in List-I to the correct entry in List-II and choose the correct option.

List-I

(P) If α and β are the distinct roots of the equation $x^2+x+1=0$, then the quadratic equation with roots $\frac{1}{(\alpha+1)^{2026}}$ and $\frac{1}{(\beta+1)^{2026}}$ is

(Q) If α and β are the distinct roots of the equation $x^2+x+1=0$, then the quadratic equation with roots $\frac{1}{(\alpha+1)^{2027}}$ and $\frac{1}{(\beta+1)^{2027}}$ is

(R) If γ and δ are the distinct roots of the equation $x^2-x+1=0$, then the value of $\frac{1}{(\gamma-1)^{2026}}+\frac{1}{(\delta-1)^{2026}}$ is

(S) If p and r are the distinct roots of the equation $x^2+x-1=0$, then the value of $\frac{1}{(p+1)^3}+\frac{1}{(r+1)^3}$ is

List-II

(1) $x^2+x+1=0$

(2) $x^2-x+1=0$

(3) $x^2+x-1=0$

(4) -1

(5) -4

(A)	(P) → (1), (Q) → (2), (R) → (5), (S) → (4)
(B)	(P) → (3), (Q) → (1), (R) → (4), (S) → (5)
(C)	(P) → (1), (Q) → (2), (R) → (4), (S) → (5)
(D)	(P) → (2), (Q) → (3), (R) → (5), (S) → (4)

Q.14

Match each entry in List-I to the correct entry in List-II and choose the correct option.

List-I	**List-II**
(P) The number of elements in the set $\{x \in [-\pi, \pi] : \sin^6 x + \cos^4 x = 1\}$	(1) is 1
(Q) The number of elements in the set $\left\{x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] : \sin^2 x + \cos^6 x = 1\right\}$	(2) is 2
(R) The number of elements in the set $\left\{x \in [-\pi, \pi] : \cos^2\left(\frac{x}{2}\right) - \sin^2 x = \frac{1}{2}\right\}$	(3) is 3
(S) The number of elements in the set $\left\{x \in [-2\pi, 2\pi] : 6\sin^2\left(\frac{x}{2}\right) - \cos 3x = 3\right\}$	(4) is 4
	(5) is 5

(A)	(P) → (2), (Q) → (5), (R) → (3), (S) → (4)
(B)	(P) → (5), (Q) → (3), (R) → (2), (S) → (4)
(C)	(P) → (5), (Q) → (4), (R) → (1), (S) → (3)
(D)	(P) → (4), (Q) → (3), (R) → (2), (S) → (5)

Q.15

For real numbers $\alpha,\ \beta,\ \gamma,\ \delta$ and μ, consider the matrix

$$M = \begin{bmatrix} \alpha & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \beta & \frac{1}{\sqrt{3}} \\ \gamma & \delta & \mu \end{bmatrix}.$$

Suppose that $MM^T = I$, where M^T is the transpose of the matrix M, and I is the 3×3 identity matrix. Let

$$\vec{u} = \alpha\,\hat{\imath} + \frac{1}{\sqrt{3}}\hat{\jmath} + \gamma\,\hat{k}, \qquad \vec{v} = \frac{1}{\sqrt{2}}\hat{\imath} + \beta\hat{\jmath} + \delta\hat{k} \quad \text{and} \quad \vec{w} = -\frac{1}{\sqrt{2}}\hat{\imath} + \frac{1}{\sqrt{3}}\hat{\jmath} + \mu\hat{k}.$$

Match each entry in List-I to the correct entry in List-II and choose the correct option.

List-I	**List-II**
(P) The value of $\gamma^2 + \delta^2$ is	(1) 0
(Q) If $x\vec{u} + y\vec{v} + z\vec{w} = \hat{\jmath}$ for some real numbers $x,\ y$ and z, then the value of x is	(2) 1
(R) The value of $\lvert\vec{u} \cdot (\vec{v} \times \vec{w})\rvert$ is	(3) $\frac{1}{\sqrt{2}}$
(S) The value of $\lvert\vec{u} \times (\vec{v} \times \vec{w})\rvert$ is	(4) $\frac{1}{\sqrt{3}}$
	(5) $\frac{5}{6}$

(A)	(P) → (5), (Q) → (4), (R) → (2), (S) → (1)
(B)	(P) → (4), (Q) → (5), (R) → (1), (S) → (2)
(C)	(P) → (5), (Q) → (3), (R) → (2), (S) → (1)
(D)	(P) → (5), (Q) → (4), (R) → (1), (S) → (2)

Q.16

Match each entry in List-I to the correct entry in List-II and choose the correct option.

List-I	**List-II**
(P) The circle with centre $(1,2)$ and touching the straight line $3x+4y=1$, passes through	(1) the point $(1,1)$
(Q) The common tangent to the circle $x^2+y^2=2$ and the parabola $y^2=8x$ with positive slope, passes through	(2) the point $(7,9)$
(R) Let M be the end point of the latus rectum of the ellipse $3x^2+4y^2=48$ such that M lies in the first quadrant. Then the normal to the ellipse drawn at M passes through	(3) the point $(3,2)$
(S) Let H be the hyperbola whose centre is at the origin, one of the foci is at $(5,0)$, and one directrix is $5x+16=0$. Then H passes through	(4) the point $(2,5)$
	(5) the point $(8,3\sqrt{3})$

(A)	(P) → (3), (Q) → (4), (R) → (1), (S) → (2)
(B)	(P) → (3), (Q) → (2), (R) → (1), (S) → (5)
(C)	(P) → (3), (Q) → (2), (R) → (4), (S) → (5)
(D)	(P) → (4), (Q) → (1), (R) → (2), (S) → (3)

END OF THE QUESTION PAPER